
FinRegQA-EU: Corruption-Based Preference Data for Grounded EU Financial Regulatory Question Answering

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Abstract

Large Language Models (LLMs) struggle with region-specific factual knowledge, particularly in financial regulation. While benchmarks such as CFinBench [Ding et al.] provide broad coverage of financial knowledge in other regions, no comparable resource exists for European financial regulation. We close this gap by proposing an end-to-end pipeline for evaluating and improving LLMs on European financial regulatory question answering. Our dataset is grounded in the official Q&A corpora of the European Banking Authority (EBA) and the European Securities and Markets Authority (ESMA). To measure current model performance, we develop a pointwise LLM-as-a-Judge evaluation using multiple judge models, and we convert the resulting judge scores into preference pairs. Dispreferred answers are synthesised through a taxonomy of regulatory failure modes, yielding pairs with clearer separation between correct and incorrect regulatory reasoning. We then fine-tune Qwen3.5-4B on 359 preference pairs with DPO improves rule-based Citation F1 across all five corruption strategies (up to +0.030), while pointwise judge scores are unchanged. This expose both the fragility of current models on European financial regulation and the blind spot of standard LLM-as-a-Judge evaluation on citation faithfulness, and showing that adversarially constructed preference data offers a practical path to correcting them.

1 Introduction

Large Language Models (LLMs) are increasingly deployed across the financial sector, including in regulator-facing workflows where outputs must align with the governing regulatory text. Existing evaluations, however, focus largely on textbook-style question answering and numerical problem solving. Chen et al. [a] target deep questions over financial documents requiring complex numerical reasoning, and Chen et al. [b] extend this to conversational, multi-turn settings. Related benchmarks [Islam et al., Wang et al., a] likewise centre on extraction and arithmetic skills. Legal benchmarks [Cao et al., Guha et al., Li et al.] address the legal domain at a generic level, but do not isolate financial regulation.

Most recently, CNFinBench [Ding et al.] closes part of this gap by evaluating LLMs along capability, compliance, and safety axes, and by introducing multi-turn adversarial tasks. Its results show that most systems achieve only partial resistance under adversarial testing, indicating that current models remain unreliable without cited, verifiable reasoning. CNFinBench, however, is built entirely around the Chinese regulatory regime. No comparable resource exists for Europe, despite the fact that the EU publishes an extensive body of open regulatory Q&A material through the European Banking Authority (EBA) and the European Securities and Markets Authority (ESMA).

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We close this gap by proposing an end-to-end pipeline for evaluating and improving LLMs on EU financial regulatory question answering, grounded in the official Q&A corpora of the EBA and ESMA. To measure how well current models answer regulatory questions, we use a pointwise LLM-as-a-Judge protocol that scores answers along four dimensions: *accuracy* against the official answer, *completeness* of key points, *topic coherence* with the stated subject matter, and *citation quality* in terms of the specificity and correctness of legal references. We employ multiple judge models, evaluate each independently, and use the resulting scores to construct preferred and less-preferred answer pairs. To sharpen the separation between the two, we synthesise less-preferred answers through a taxonomy of regulatory failure modes including article swap, law swap, and hallucination injection. Because our aim is to improve models on their own output distribution, we fine-tune on these pairs and evaluate against the base model.

Contributions.

- **An EU regulatory QA evaluation benchmark** built from authoritative EBA and ESMA Q&A material, scored along four interpretable dimensions: accuracy, completeness, topic coherence, and citation quality.
- **A regulatory failure-mode taxonomy** including article swap, law swap, and hallucination injection for synthesising calibrated less-preferred answers, yielding preference-learning datasets that target the specific errors LLMs make in this regime.
- **A controlled comparison** of preference-optimisation methods Standard DPO for fine-tuning small open-weight LLMs, evaluated on citation F1 against the official answer and preference accuracy.

2 Related Work

Legal and financial QA benchmarks. Most financial benchmarks test numerical reasoning: reading tables, extracting numbers, doing arithmetic [Chen et al., a,b, Islam et al., Wang et al., a]. Legal benchmarks [Guha et al., Li et al., Cao et al.] cover law in general, but financial regulation is only a small part of what they test. The closest work to ours is CNFinBench [Ding et al.], which does focus on financial regulatory compliance but only for China. Because regulation is jurisdiction-specific by constructions and currently, no equivalent resource exists for European Union. We close this gap by introducing a benchmark for EU Financial Regulatory QA grounded in the official Q&A material published by the EBA and ESMA.

LLM-as-a-Judge. Using one LLM to score another model’s answers is now standard practice, and prior work reports that judges agree with humans about as well as humans agree with each other [Zheng et al., Licht et al., Wang et al., b]. That work also documents the method’s failure in position bias (favoring whichever answer appears first), verbosity bias, and self-preference [Zheng et al., Zeng et al.]. These biases affect pairwise comparison in particular, which is one reason we score answers one at a time rather than in pairs. Pointwise scoring also gives us something pairwise does not which is a scalar score per dimension that can be the threshold for building preference pairs.

Citation grounding and attribution. A common LLMs failure is citations hallucination. A model can produce a fluent answer and attach a citation that does not support it. This problem has been studied under some past research, Liu et al. [a] formalize what it means for a statement to be supported by a source, and Gao et al. build a benchmark for text generation with citations. The finding most relevant to us is that retrieval does not solve the problem. Liu et al. [b] also show that commercial generative search engines which retrieve documents before answering still frequently produce citations that do not back up their claims. Onweller et al. report the same pattern for frontier models, with only 39–77% of cited claims verified correct, and find that adding retrieval does not close the gap. Similar issues appear in academic citation verification [Sadeghi et al.].

Thus, it is clear that if more retrieval does not produce better citations, then by fixing the model, we can get better output compared to fixing the pipeline. That is why we fine-tune the model to produce a better citation-grounded output.

Preference optimisation. Direct Preference Optimization (DPO) trains a model directly on pairs of preferred and rejected responses, without a separate reward model [Rafailov et al.]. This method

has been used mainly to align open-weight models. DPO learns little when rejected answer is never produced by the model, since the gradient pushes probability away. Just Train Twice (JTT) upweights the examples a model gets wrong on a first pass [Liu et al., a] and DORO improves worst-case performance across groups [Zhai et al.] . We compare these against standard DPO since regulatory QA has the most error on fabricated citations.

The closest work to ours is Ovcharov, who use DPO to reduce citation hallucination by corrupting correct citations to build rejected answers. But they corrupt human-written reference answers. The resulting rejected answers are directly from ground-truth, which is off-policy method. In this research, we corrupt the model’s own answers instead, selected by judge score, which means we train against the mistakes the model actually makes.

3 Datasets

Sources We construct our evaluation corpus from the two official Q&A repositories maintained by the European Banking Authority (EBA) [European Banking Authority] and the European Securities and Markets Authority (ESMA) [European Securities and Markets Authority]. Both regulators operate public single-rulebook Q&A platforms in which supervised entities submit interpretive questions on specific provisions of EU financial regulation, and the regulator (or the relevant joint committee) issues an authoritative written answer. These answers functions as legal cited supervisory for compliance practice.

The EBA Q&A covers regulatory on the practical application or implementation of the banking, payment services, AML/CFT and other legislation that falls within the EBA’s remit. This includes the associated delegated and implementing acts, RTS, ITS, guidelines and recommendations. While ESMA Q&A covers securities market and aim to ensure a consistent application of the EU Single Rulebook for financial services. Together they provide a controlled, expert-curated source of regulatory question-answer pairs that is, to our knowledge, the most direct ground-truth signal available for EU financial regulatory QA.

Schema and Preprocessing Each raw entry on both portals provides a question, an answer, and structured metadata indicating the legal act under interpretation, the topic area, and the subject matter and background of the question. We parse each entry into a uniform schema with six fields: legal act, topic, subject matter, background, question and ground truth. The legal act field is normalised to its official CELEX-style identifier when available (e.g., Regulation (EU) 2022/2554 for DORA). The topic label is retained as published by the regulator, since it reflects the supervisor’s own taxonomy.

Table 1: Dataset composition. Each source question is answered by two answerer models and scored independently by three judges.

Statistic	Count
Source questions	2,138
from EBA	1,868
from ESMA	270

Preference-Pair Construction For preference learning we construct a corpus of preferred / rejected answer pairs from the training split. Candidate answers are generated by two open-weight 70B answerer models, there are Apertus-70B- Instruct and Llama-3.3-70B-Instruct, each conditioned on the structured context fields using the prompt in Appendix B.1. Every candidate is scored by the three-judge ensemble along *accuracy*, *completeness*, *topic coherence*, and *citation quality*; the four scores are combined into a weighted quality score $s \in [1, 5]$ with higher weight on accuracy and citation quality, the two dimensions most predictive of regulatory utility.

A candidate enters the *preferred* set if its weighted score satisfies $s \geq \tau_{\text{pref}}$. To produce a corresponding *less-preferred* answer for each preferred candidate, we apply one of four targeted corruption operations, each instantiating a distinct failure mode observed in baseline answerer outputs:

- **Article swap.** A cited article within the correct legal act is replaced with a plausible but incorrect article from the same act.

- **Law swap.** The legal act itself is replaced with an adjacent but incorrect act (e.g., MiFID II $\leftrightarrow$ MiFIR).
- **Anachronism.** A correct citation is replaced with a superseded or not-yet-applicable version of the provision (e.g., a pre-amendment CRR article number, or an act not yet in force at the question date).
- **Hallucination injection.** A fabricated but plausible-sounding claim (a non-existent article, an invented numeric threshold, or a made-up implementing decision) is inserted into an otherwise correct answer.

To ensure that each pair is distinguishable yet non-trivial, we retain a pair only when the score gap exceeds a separability threshold $s_{\text{pref}} - s_{\text{neg}} \geq \tau_{\text{sep}}$, discarding pairs in which the corruption was too subtle for the judge to detect or in which the original preferred candidate was itself borderline.

4 Methods

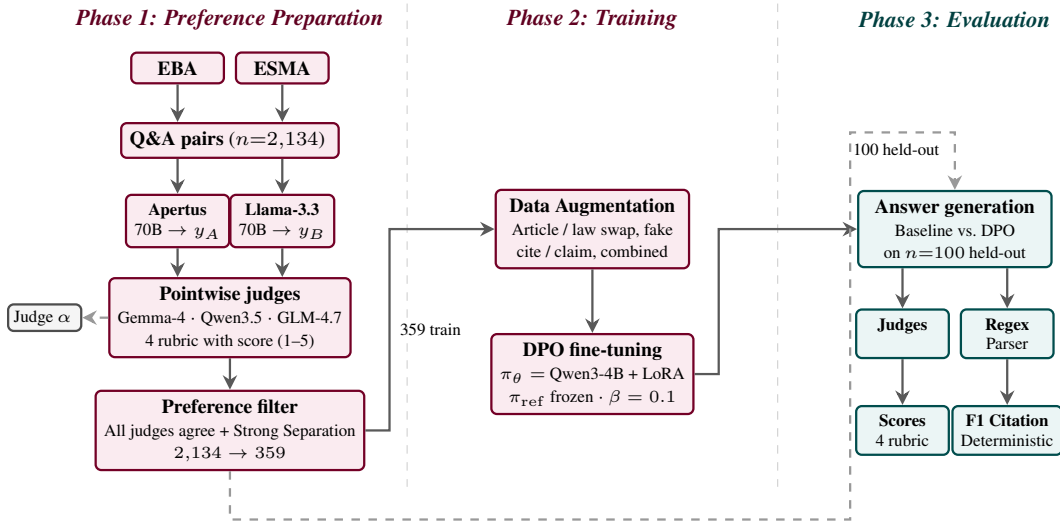


Figure 1: End-to-end EU-Reg-QA framework. **Phase 1** constructs preference pairs from the EBA and ESMA regulatory Q&A corpora: two 70B answerer models generate candidate answers y_A, y_B ; a pointwise ensemble of three judges scores each candidate on four unan dimensions (accuracy, completeness, coherence, citation quality); a unanimity plus strong-preference filter yields 359 training pairs and 100 held-out evaluation pairs; inter-judge Krippendorff α is computed as a side diagnostic. **Phase 2** augments the training set with five corruption strategies (article swap, law swap, fabricated citation, fabricated claim, and combined) and fine-tunes Qwen3.5-4B (LoRA) with DPO against a frozen reference. **Phase 3** evaluates the baseline and DPO checkpoint on the 100 held-out questions using the same pointwise judge ensemble (judge scores and pairwise win rate) and a rule-based Citation F1 metric at both instrument and article granularity.

We evaluate and improve LLM performance on EU financial regulatory question answering (QA) in three stages: (i) a *pointwise LLM-as-a-Judge* protocol that scores candidate answers along four rubric dimensions; (ii) *preference-based fine-tuning*, in which judge scores are converted into preference pairs and further augmented with targeted corruptions; and (iii) an *evaluation strategy* that combines the same judge ensemble with a rule-based Citation F1 metric to compensate for the judges’ known weakness on citation quality.

4.1 Pointwise Prompting with LLMs

We adopt a pointwise *LLM-as-a-Judge* protocol rather than a pairwise one. Pointwise scoring is known to be more reliable than pairwise comparison for measuring scalar constructs and avoids the

position bias documented in pairwise setups [Licht et al., Zeng et al.]; the trade-off is that pointwise judges are more vulnerable to adversarial responses [Raina et al.], which we exploit deliberately in Section 4.2.

Rubric. Each candidate answer a to a regulatory question q (drawn from the EBA and ESMA Q&A corpora, with the official answer a^* as reference) is scored on a 1–5 Likert scale along four dimensions: **Accuracy**, **Completeness**, **Topic Coherence**, and **Citation Quality**. Each dimension is anchored by explicit level descriptors (see Appendix B.2); the overall score is the arithmetic mean of the four.

Judge ensemble. To reduce single-model bias, every (q, a, a^*) triple is scored independently by three open-weight judges: *Gemma-4-31B-it*, *Qwen3.5-27B*, and *GLM-4.7-Flash*. Judges are prompted with the rubric, the question, the reference answer, and the candidate answer, and are required to return a JSON object with one sentence of reasoning per dimension followed by an integer score. To avoid truncation inside reasoning blocks, we set `max_new_tokens = 2048` and disable extended thinking mode where supported. The final pointwise score for each dimension is the mean across the three judges.

Preference derivation. For every question, each of the three judges scores both candidate answers (from *Apertus-70B-Instruct* and *Llama-3.3-70B-Instruct*) independently. For every (question, judge) pair we compute an *overall* score as the unweighted mean of the four rubric dimensions (Accuracy, Completeness, Topic Coherence, Citation Quality), and record which answerer that judge prefers. A pair is retained as a preference example only if two conditions hold:

- (i) **Unanimous agreement** : all three judges prefer the same answer (majority count = 3);
- (ii) **Strong preference** : at least one judge exhibits a score gap of ≥ 1.5 between the two answers.

The retained answer is labelled *chosen* and the other *rejected*; all remaining pairs are discarded to reduce label noise from ambiguous or judge-specific preferences.

Out of $N_{\text{total}} = 2,134$ questions, 858 (40.2%) yield unanimous judge agreement, of which 359 additionally satisfy the strong-preference criterion. We use these 359 pairs as the DPO training set and hold out an untouched set of 100 pairs for evaluation.

Table 2: Preference pair construction. For each question, judges score both answerer models on an overall score (mean of the four dimensions). A pair is retained only if all three judges agree on the preferred answer and at least one judge shows a strong preference (score gap ≥ 1.5).

Criterion	Questions	%
Total questions	2,134	100.0
Unanimous (all available judges agree)	913	42.8
Unanimous (all 3 judges, majority = 3)	858	40.2
Unanimous <i>and</i> strong preference	359	16.8

Table 3: Composition of DPO-ready pairs ($n = 359$), by regulator and preferred model.

Regulator	Preferred model		Total
	Llama-3.3-70B	Apertus-70B	
EBA	232	64	296
ESMA	54	9	63
Total	286	73	359

4.2 Fine-tuning

Base model. We fine-tune QWEN3.5-4B as the student model. The 4B scale is small enough to iterate over the full corruption grid on a single GH200 GPU while remaining strong enough that gains reflect the *method*, not the base model.

DPO objective. We use Direct Preference Optimization [Rafailov et al.], which optimizes the policy π_θ against a frozen reference π_{ref} (the base Qwen3.5-4B) using the standard log-sigmoid loss on preference pairs (q, y^+, y^-) :

$$\mathcal{L}_{\text{DPO}} = -\mathbb{E}_{(q, y^+, y^-)} \log \sigma \left(\beta \left(\log \frac{\pi_\theta(y^+|q)}{\pi_{\text{ref}}(y^+|q)} - \log \frac{\pi_\theta(y^-|q)}{\pi_{\text{ref}}(y^-|q)} \right) \right). \quad (1)$$

Training uses LoRA adapters (rank $r = 16$, $\alpha = 32$) on all attention and MLP projections with 4-bit base weights, `learning_rate` = 5×10^{-6} , $\beta = 0.1$, effective batch size 16, and 3 epochs. Checkpoints are selected on validation overall score.

Corruption strategies. Beyond the natural preference pairs above, we construct four targeted *corrupted* training sets by keeping the chosen answer y^+ intact and replacing y^- with a systematically degraded version of it. Each strategy targets a specific failure mode observed in the baseline outputs. Each corruption defines an independent training run; we also report results on the natural pairs alone as a baseline.

4.3 Evaluation Strategy

We evaluate every fine-tuned checkpoint on a held-out test split of $n = 100$ EU regulatory questions, disjoint from the training and validation splits. For each question we generate one answer from the baseline (Qwen3.5-4B) and one from the DPO checkpoint, then report three families of metrics.

Judge scores. The same pointwise judge ensemble described in Section 4.1 scores each answer along the four rubric dimensions. We report the mean across judges for each dimension and the overall score.

Pairwise win rate. For each question, we compare the mean overall scores of the baseline and DPO answers and record whether DPO is preferred, baseline is preferred, or the two tie (equal to two decimal places).

Rule-based citation metric. Judge agreement collapses on Citation Quality (Krippendorff’s $\alpha \approx 0.01$ – 0.41) despite strong agreement on Accuracy ($\alpha \approx 0.68$ – 0.78 ; see Section 5). We therefore complement the judge ensemble with a rule-based Citation F1 metric described next.

4.3.1 Evaluation Metrics: Citation F1

Citation F1 treats each answer as a set of regulatory references and compares it to the reference set extracted from the official answer a^* . References are extracted with a deterministic regular-expression parser that recognises the instrument name (e.g., *CRR*, *MiFIR*, *DORA*, Regulation (EU) 2019/876) and, where present, an article, paragraph, sub-paragraph and point identifier.

We report Citation F1 at two granularities:

- **Instrument-level Citation F1:** the citation is considered correct if the referenced regulation matches a^* , regardless of article. This captures whether the model cites the right *law*.
- **Article-level Citation F1:** the citation is considered correct only if both the instrument and the article identifier match. This captures whether the model cites the right *provision*.

For an answer a with predicted citation set $C(a)$ and gold citation set $C(a^*)$, precision, recall and F1 are computed in the standard way:

$$P = \frac{|C(a) \cap C(a^*)|}{|C(a)|}, \quad R = \frac{|C(a) \cap C(a^*)|}{|C(a^*)|}, \quad F_1 = \frac{2PR}{P + R}. \quad (2)$$

Compared with judge-based Citation Quality, this metric is deterministic, cheap to compute, and directly discriminates *fabricated* citations from *correct* ones. In Section 5 we show that it is also more discriminative across models than either the judges’ Citation Quality dimension or a cosine-similarity baseline.

5 Results

Pointwise judge results. Both answerer models score poorly on Accuracy (2.13–2.40) and Completeness (1.86–2.32) while scoring consistently high on Topic Coherence (above 4.0). The pattern is similar, answers stay on topic and read fluently, but the substance is often wrong or incomplete relative to the regulator’s reference answer. Results are broadly consistent across the EBA and ESMA subsets. While EBA scores are marginally higher on every dimension, but the gap is small enough to suggest that the difficulty of the two corpora is comparable.

Accuracy and Completeness are the axes on which LLMs struggle. Topic Coherence is easy for the answerer models, where both stay comfortably above 4.0 on the 1–5 scale for both regulators. Accuracy and Completeness, by contrast, sit around 2, indicating that the models rarely reproduce the regulator’s answer in full and often misstate its substance. This gap between surface fluency and factual correctness is the failure mode we target with the preference-based fine-tuning of §4.2.

Apertus-70B underperforms Llama-3.3-70B despite matched scale. At the same 70B parameter count, Apertus consistently underperforms Llama-3.3 on every rubric dimension, with per-metric gaps of 0.2–0.5 points. We interpret this as an instruction-tuning effect rather than a capacity effect and note it as evidence that parameter count alone is a poor proxy for competence in specialized regulatory QA.

Table 4: Mean judge scores (1–5) by answerer model and regulator, averaged across the three judges.

Regulator	Model	Accuracy	Completeness	Coherence	Citation
EBA	Llama-3.3-70B	2.40	2.32	4.57	3.29
	Apertus-70B	2.13	2.12	4.38	2.89
ESMA	Llama-3.3-70B	2.27	2.11	4.47	3.14
	Apertus-70B	1.94	1.86	4.12	2.48

Inter-judge agreement is high on Accuracy and Completeness, but collapses on Citation Quality. Judges reach substantial agreement on Accuracy and Completeness, but agreement drops sharply on Topic Coherence and, most severely, on Citation Quality. This asymmetry motivates the design of our evaluation stack: we complement the judge ensemble with a deterministic rule-based Citation F1 metric (§4.3.1) so that the axis on which judges are least reliable is precisely the axis on which our reported gains do not depend on the judges.

Table 5: Pairwise inter-judge agreement (Krippendorff’s α). Judges agree substantially on accuracy and completeness, but agreement collapses on citation quality.

Dimension	Judge pair		
	Qwen–Gemma	Qwen–GLM	Gemma–GLM
Accuracy	0.781	0.682	0.738
Completeness	0.737	0.695	0.703
Topic coherence	0.558	0.455	0.399
Citation quality	0.345	0.014	0.413

Preference derivation. From the pointwise judge scores, we derive preference pairs between the two answerer models per the criteria of §4.1. As expected, the largest chosen–rejected gaps occur on Accuracy and Completeness, the same dimensions on which the models struggle most. While ties are more frequent on Topic Coherence, where both models score highly. This confirms that the preference signal is concentrated on the axes we most want the fine-tuned model to improve.

Table 6: Mean absolute score gap $|s_{\text{chosen}} - s_{\text{rejected}}|$ per dimension, over DPO-ready pairs. Accuracy drives the preference signal, topic coherence contributes least.

Dimension	Mean gap (1–5)
Accuracy	1.998
Completeness	1.461
Citation quality	1.218
Topic coherence	0.928

Fine-tuning results. Across all five corruption strategies, DPO produces only marginal changes in aggregate judge scores: overall scores shift by at most $+0.037$ on the 1–5 scale (Table 7), and pairwise win rates hover near 50/50 (Table 8). Taken at face value, these numbers would suggest that corruption-based DPO has little effect on answer quality based on the judge score.

However, the rule based citation F1 metric tells a different result, all five corruption strategies improve over the baseline, with gains of up to $+0.030$ ($\approx 13\%$ relative). The "combined" corruption strategy, which introduces the combination of failure modes in rejected samples, gain the highest citation F1 improvement.

The pattern of the improvement of F1 citation metrics, contradicts with the accuracy score from the judge. Combined and Article Swap achieve the highest Citation F1 gains but the lowest judge Accuracy deltas (-0.104 and -0.050 respectively), while Law Swap, which achieves the second *lowest* Citation F1 gain, achieves the *highest* judge Accuracy delta (-0.010 , the smallest regression). This ranking reversal is not an independent case, but it is systematic across the five strategies.

The disagreement is consistent with the inter-judge agreement analysis of Table 5. Judges reach substantial agreement on Accuracy (Krippendorff’s $\alpha \approx 0.68$ – 0.78) but lower agreement on Citation Quality ($\alpha \approx 0.01$ – 0.41). In other words, the axis on which DPO produces the largest and most consistent improvements is precisely the axis on which the judge ensemble is least reliable.

Table 7: DPO fine-tuning results on Qwen3.5-4B across five preference-pair corruption strategies: pointwise judge scores on the four rubric dimensions. Each cell reports the DPO score with the change relative to the strategy-specific baseline in parentheses. Judge ensemble: Gemma-4-31B-it, Qwen3.5-27B, GLM-4.7-Flash. Scores use a 1–5 scale, averaged over 100 held-out questions. **Overall** is the mean of the four rubric dimensions.

Corruption	Accuracy	Completeness	Coherence	Cite Quality	Overall
Hallucinate Citation	1.973 $-.040$	2.083 $+.050$	4.203 $+.040$	2.597 $-.010$	2.714 $+.010$
Article Swap	1.970 $-.050$	2.010 $-.054$	4.243 $+.100$	2.660 $+.055$	2.721 $+.013$
Hallucinate Text	1.970 $-.040$	2.033 $-.067$	4.214 $+.091$	2.569 $-.055$	2.696 $-.018$
Law Swap	2.003 $-.010$	2.070 $+.014$	4.184 $+.047$	2.709 $+.100$	2.742 $+.037$
Combined (all four)	1.906 $-.104$	1.997 $-.060$	4.207 $+.090$	2.592 $-.040$	2.676 $-.029$

Table 8: DPO fine-tuning results on Qwen3.5-4B: pairwise judge preference and rule-based Citation F1. Win/Loss/Tie is the DPO-preferred / baseline-preferred / tie rate over 100 held-out questions from the same judge ensemble as Table 7. Citation F1 is computed at instrument level from a deterministic regex parser; the baseline Citation F1 is 0.228 for all strategies.

Corruption	Win / Loss / Tie (%)	Cite F1	Δ Cite F1
Hallucinate Citation	44 / 46 / 10	0.239	$+.011$
Article Swap	42 / 48 / 10	0.254	$+.026$
Hallucinate Text	47 / 46 / 7	0.244	$+.017$
Law Swap	46 / 40 / 14	0.242	$+.014$
Combined (all four)	42 / 42 / 16	0.258	$+.030$

6 Conclusion

We introduce a novel benchmark for financial regulatory question answering, a domain where correctness and precise citation of legal instruments are non-negotiable. A fabricated article number or a swapped regulation can carry real compliance consequences, and any deployed system must be evaluated against that bar. Our results show that even 70B-scale open-weight LLMs (Apertus-70B, Llama-3.3-70B) struggle to produce factually accurate answers with complete and correctly attributed citations to the underlying EU regulatory instruments. Fine-tuning a Qwen3.5-4B student with DPO improves citation faithfulness across all five corruption strategies (Citation F1 up to +0.030), while pointwise judges produces only marginal changes on the same held-out set. Based on the failure modes, reveals that *accuracy is the most fragile dimension*, it decline under every corruption strategy, whereas *citation quality is the most tractable*, responding consistently to targeted preference training. Taken together, we release the benchmark and evaluation stack to enable more careful, citation-aware evaluation of any LLM proposed for compliance use.

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A Datasets

A.1 Dataset Generation Result Statistics

Table 9: Pairwise inter-judge agreement. κ_{lin} and κ_{quad} are linear- and quadratic-weighted Cohen’s κ ; α is Krippendorff’s alpha.

Dimension	Judge pair	n	κ_{lin}	κ_{quad}	α
Accuracy	Qwen–Gemma	4,276	0.648	0.785	0.781
	Qwen–GLM	4,183	0.548	0.692	0.682
	Gemma–GLM	4,183	0.596	0.738	0.738
Completeness	Qwen–Gemma	4,276	0.631	0.740	0.737
	Qwen–GLM	4,183	0.547	0.696	0.695
	Gemma–GLM	4,183	0.552	0.704	0.703
Topic coherence	Qwen–Gemma	4,276	0.426	0.559	0.558
	Qwen–GLM	4,183	0.341	0.458	0.455
	Gemma–GLM	4,183	0.305	0.405	0.399
Citation quality	Qwen–Gemma	4,276	0.277	0.432	0.345
	Qwen–GLM	4,183	0.130	0.239	0.014
	Gemma–GLM	4,183	0.327	0.435	0.413

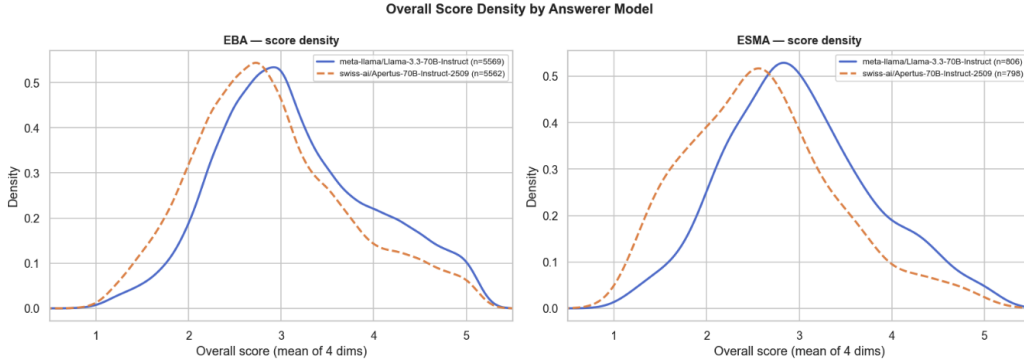


Figure 2: Score Distribution across answerer model

Table 10: Correlation between embedding cosine similarity (all-mpnet-base-v2, answer vs. official answer) and judge scores. Despite the correlations are near zero across all dimensions, the similarity got average (0.777) across all answers vs ground truth. Indicating that the semantic similarity also agree on what judges measure. $n = 12,711$.

Dimension	Pearson r	Spearman ρ
Accuracy	0.058	0.072
Completeness	0.078	0.112
Topic coherence	0.090	0.089
Citation quality	0.076	0.089

B Prompts

This appendix contains the full text of the prompts used in our pipeline. Placeholders enclosed in curly braces (e.g. {legal_act}) are filled at inference time with the corresponding fields of the evaluation instance.

Table 11: Zero-shot topic matching against judge *topic coherence* scores. Answers are embedded and compared to all 154 ground-truth topic labels; `sim_to_gt_topic` is the cosine similarity to the correct label, and `topic_margin` is that similarity minus the highest similarity to any other label (positive = correctly on-topic). Both correlate negatively and near-zero with the judge scores.

Metric	Mean	Min	Max	Pearson r	Spearman ρ
<code>sim_to_gt_topic</code>	0.323	-0.074	0.743	-0.078	-0.092
<code>topic_margin</code>	-0.107	-0.583	0.372	-0.076	-0.091

Table 12: DPO fine-tuning of Qwen3.5-4B (LoRA, 4-bit, $\beta = 0.05$, 3 epochs) on combined-corruption pairs. Judge scores decrease slightly; win rate is at chance; only Citation F1 improves. $n = 100$ evaluation questions.

Metric	Baseline	DPO	Δ
<i>Judge scores (1–5)</i>			
Accuracy	2.014	1.875	-0.139
Completeness	2.062	1.972	-0.090
Topic coherence	4.128	4.184	+0.056
Citation quality	2.610	2.573	-0.037
Overall (mean)	2.703	2.651	-0.052
<i>Pairwise win rate</i>			
DPO preferred	43.0%		
Baseline preferred	44.0%		
Tie	13.0%		
<i>Rule-based</i>			
Citation F1	0.228	0.258	+0.030

B.1 Answer Generation Prompt

The answer generation prompt elicits an EBA/ESMA-style response to a regulatory question conditioned on the structured context of the instance (legal act, topic, subject matter, and background).

```
You are an expert in EU financial regulation, with deep knowledge of EBA and ESMA
guidelines, technical standards, and related directives and regulations.

You will receive a regulatory question along with structured context:
- LEGAL ACT: the specific instrument the question concerns
- TOPIC: the regulatory area
- SUBJECT MATTER: the specific provision, field, or requirement
- BACKGROUND: the context or inconsistency motivating the question

## Requirements

1. **ACCURACY**: Base your answer on the actual content of the legal act identified above
and any directly relevant secondary instruments (RTS, ITS, guidelines). Address the specific
subject matter and background; do not answer a more general version of the question.

2. **CITATIONS**: Every substantive claim must be supported by a specific citation in the
form [Source, Article/Paragraph/Field], e.g., [Regulation (EU) 2022/2554, Art. 28(3)] or
[EBA/ITS/2023/01, field B_05.01.0020]. The primary citation should point to the legal act
named in the context. General references such as "DORA" or "the ITS" are not sufficient. Use
square brackets consistently.

3. **ABSTENTION**: If you cannot identify a specific provision for a claim, write "[no
specific provision identified]" immediately after the claim, rather than fabricating a
citation. Partial answers with honest gaps are preferred over complete answers with
fabricated references.
```

4. ****FORMAT****: 100-400 words of prose, matching the style of official EBA/ESMA Q&A responses. Inline citations after each substantive claim. Do not pad.

5. ****CONCISENESS****: State each point once. Do not restate the question, do not write a closing "In summary" / "In conclusion" paragraph that repeats what you already said, and do not hedge with filler like "it is worth noting" or "however, it should be noted". If you do not know something, say so plainly in one sentence rather than reasoning around it at length.

Context

LEGAL ACT: {legal_act}
 TOPIC: {topic}
 SUBJECT MATTER: {subject_matter}
 BACKGROUND: {background}

Question

{question}

Answer :

B.2 Judge Prompt

The judge prompt scores a candidate answer against the official EBA/ESMA answer on four dimensions (*accuracy*, *completeness*, *topic coherence*, and *citation quality*) using a five-point rubric, and returns a structured JSON object with one reasoning sentence and one integer score per dimension.

You are an expert judge evaluating answers to EU financial regulation questions from EBA and ESMA sources. You will score a candidate answer against the official answer on four dimensions.

Question
 {question}

Topic
 {topic}

Subject Matter
 {subject_matter}

Official Answer (Ground Truth)
 {ground_truth}

Candidate Answer
 {candidate}

Scoring Dimensions

****Accuracy**** (factual correctness relative to the official answer):

- 5: All factual claims match the official answer.
- 4: Minor inaccuracies that do not change the substantive conclusion.
- 3: Partially correct; one substantive claim is wrong or unsupported.
- 2: Multiple substantive errors; conclusion is partly incorrect.
- 1: Conclusion contradicts the official answer or is fabricated.

****Completeness**** (coverage of key points in the official answer):

- 5: Covers all key points present in the official answer.
- 4: Covers all key points but omits a minor detail.
- 3: Covers the main point but misses one secondary point.
- 2: Misses multiple key points; partial coverage.
- 1: Misses the main point entirely.

****Topic Coherence**** (alignment with the specified topic and subject matter):

- 5: Fully addresses the specified topic and subject matter without drifting into adjacent regulatory areas.

- 4: Stays on topic but includes minor tangential content, OR addresses a broader scope that still fully covers the topic.
- 3: Partially on topic; some content addresses a different but related regulatory area.
- 2: Primarily addresses an adjacent topic; only partially relevant to the specified subject matter.
- 1: Off-topic or addresses a different regulatory area entirely.

****Citation Quality**** (specificity and correctness of legal references):

- 5: All citations are specific (article/paragraph/field level) and correctly identify the supporting provision.
- 4: Citations are specific and mostly correct; one minor citation issue.
- 3: Citations are present but partially generic (e.g., "Article 5" without specifying the regulation), or one citation is incorrect.
- 2: Citations are mostly generic ("the ITS", "DORA") or several are incorrect.
- 1: Citations are missing or fabricated.

Instructions

1. ****Reason before scoring.**** Think step-by-step about each dimension before assigning the score. Your reasoning should determine the score, not the reverse.
2. ****Catch plausible-sounding hallucinations.**** If the candidate contains specific factual claims (numbers, dates, article references, requirements, definitions) that are not supported by the official answer, treat those as accuracy violations even if the claims sound plausible.
3. ****Score dimensions independently.**** If two dimensions seem to conflict, score each on its own merits.
4. ****Use the full 1-5 range.**** Do not default to 3 when uncertain. A shorter candidate that still covers the key point is not a completeness penalty.
- 4b. ****Score must match your own reasoning.**** If your reasoning sentence for a dimension names a specific flaw (a missing point, a wrong citation, an unsupported claim, drift off-topic), the score for that dimension cannot be 5. Reserve 5 only when your reasoning states the candidate fully and correctly meets the criterion with no caveat. Do not default to 5 out of leniency.
5. ****Citation specificity matters.**** A correct citation to "Article 5 of Regulation (EU) 2019/2033" is stronger than "Article 5" alone, even when both are technically correct.
6. ****Be concise.**** Do your step-by-step thinking silently. Each reasoning field must be ONE short sentence (max ~25 words) stating the conclusion, not a transcript of your deliberation. Do not repeat the question, the candidate text, or the official answer back. Output the JSON object immediately after you have decided the scores -- no preamble, no text before or after the JSON.
7. ****No visible thinking.**** Do not output a `` block, chain-of-thought, or any reasoning outside the JSON's "reasoning" fields. Your entire response must be the JSON object and nothing else.

Respond in this exact JSON format only:

```
{
  "reasoning": {
    "accuracy": "<one sentence>",
    "completeness": "<one sentence>",
    "topic_coherence": "<one sentence>",
    "citation_quality": "<one sentence>"
  },
  "accuracy": <1-5>,
  "completeness": <1-5>,
  "topic_coherence": <1-5>,
  "citation_quality": <1-5>
}
```

C Corruption Strategies: Worked Example

To make the corruption strategies of Section 4.2 concrete, we illustrate each of the five perturbations on a single EBA Q&A item concerning the use of non-EUR currency equivalents for EUR thresholds under the RTS. In every case, the *chosen* answer is the reference regulator answer; only the *rejected* side is corrupted. Corrupted spans are highlighted in **red**; original spans retained from the reference answer are shown in gray.

Source item (EBA Q&A)

Question. May payment service providers (PSPs) and card schemes set rounded and easily understandable non-EUR currency equivalents for the EUR thresholds set out in the RTS?

Reference answer (excerpt). “According to *Regulation (EU) 2022/2554*, Art. 28(3), the thresholds for exemptions in the RTS are set in EUR...”

(1) **law_swap.** The article number is preserved, but the cited instrument is replaced with a topically adjacent but incorrect regulation or directive. This targets failure modes in which the model recalls a plausible article number under the wrong legal instrument.

“According to *Directive 2011/61/EU (AIFMD)*, Art. 28(3), the thresholds...”

(2) **article_swap.** The cited instrument is preserved, but the article identifier is replaced with a different article of the *same* instrument. This targets the more subtle failure mode of citing the correct law but the wrong provision.

“According to *Regulation (EU) 2022/2554*, Art. 126(3), the thresholds...”

(3) **hallucinate_citation.** A fabricated citation — one whose regulation number and article do not exist — is appended to an otherwise fluent, non-committal answer. This targets the fabricated-citation pattern observed frequently in the baseline outputs (Section ??).

“...[no specific provision identified]. This conclusion is further reinforced by *Regulation (EU) No 9138/2039, Article 731*, which explicitly mandates additional disclosure requirements not otherwise referenced here.”

(4) **hallucinate_text.** No fabricated citation is introduced; instead, a plausibly worded but factually unsupported regulatory claim is inserted. This isolates *content* hallucination from *citation* hallucination.

“...[no specific provision identified]. It should also be noted that **the relevant competent authority may, at its sole discretion, exempt branches with fewer than 50 employees from this obligation.**”

(5) **combination.** A random subset of the above four strategies is applied jointly to the same answer, producing rejected examples in which multiple failure modes co-occur.

“According to *Regulation (EU) No 1093/2010*, Art. 90(3)... This conclusion is further reinforced by *Regulation (EU) No 9829/2070, Article 841*, which explicitly mandates additional disclosure requirements not otherwise referenced here. **This obligation does not apply during the first three years following the entity’s initial authorisation.**”

Table 13 summarises which failure mode each strategy is designed to target.

Table 13: Summary of corruption strategies and the failure mode each one targets.

Strategy	Perturbation	Targeted failure mode
law_swap	Instrument → different law, article kept	Wrong law, plausible article
article_swap	Article → different article, law kept	Right law, wrong provision
hallucinate_citation	Fabricated regulation/article inserted	Non-existent citations
hallucinate_text	Fabricated regulatory claim inserted	Content hallucination without citation
combination	Random subset of the above four	Co-occurring failure modes